CLAIMS

What is claimed is:

1. A computing system, comprising:

- a plurality of computational units; and
- a massively parallel optical data fabric connecting the plurality of computational units;
- wherein at least one computational unit is configured to remain in a first power state and to enter a second, higher power state to perform a computation in response to receiving a data frame via the optical data fabric.

2. The system of claim 1, wherein the system does not include a global, high-frequency clock signal distributed to all computational units.

3. The system of claim 1, wherein computation is scheduled according to a data-flow model.

4. The system of claim 1, wherein the arrival of a data frame at a computational unit acts as an event trigger for that unit to perform an operation.

5. The system of claim 1, wherein the optical data fabric comprises a plurality of optical buses, each with a bit-width greater than 128 bits.

6. The system of claim 1, wherein the optical data fabric is an implementation of the interface described in U.S. Patent Application No. 19/559,705.